

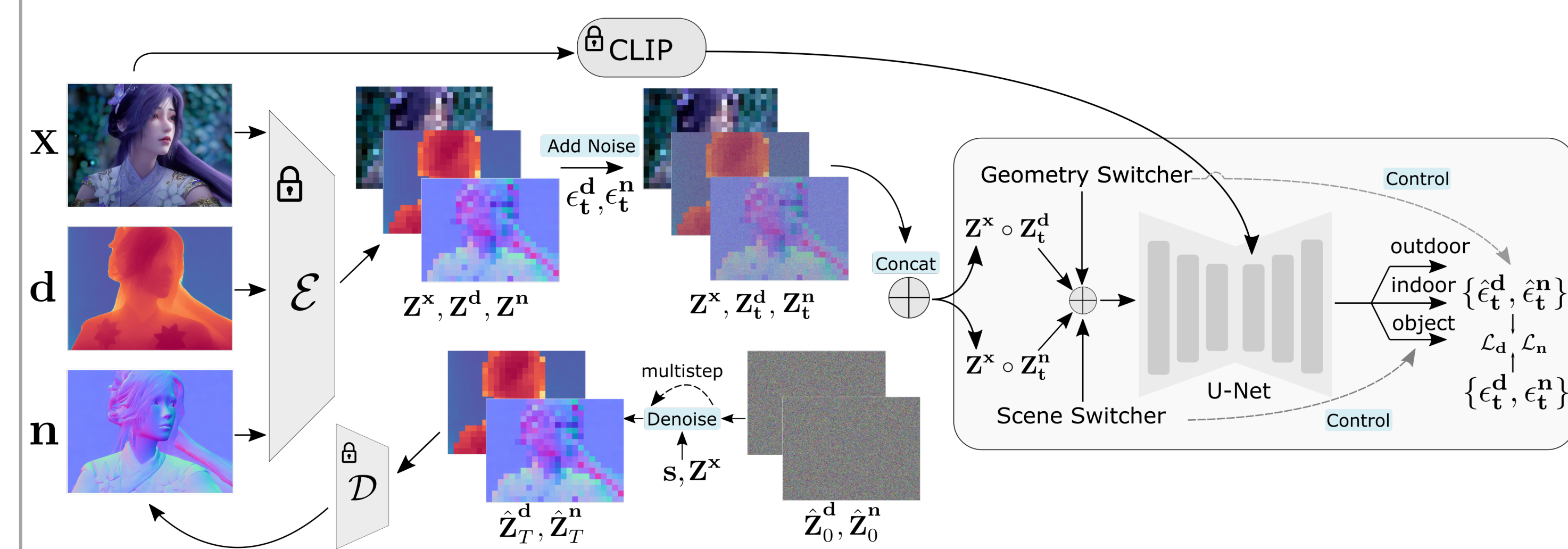
GeoWizard: Unleashing the Diffusion Priors for 3D Geometry Estimation from a Single Image

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Motivation

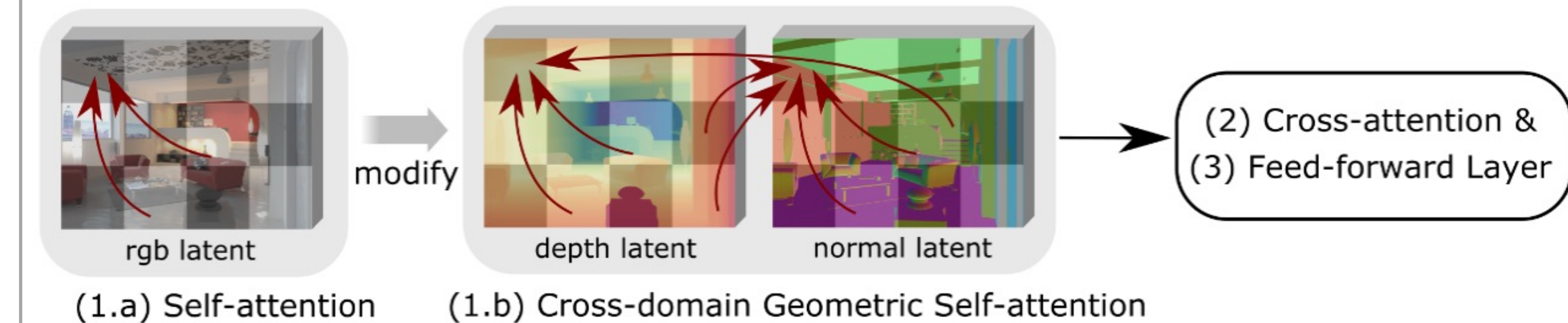
Latent diffusion models store priors to recover **high-frequency geometric details** with **limited data**.

Pipeline

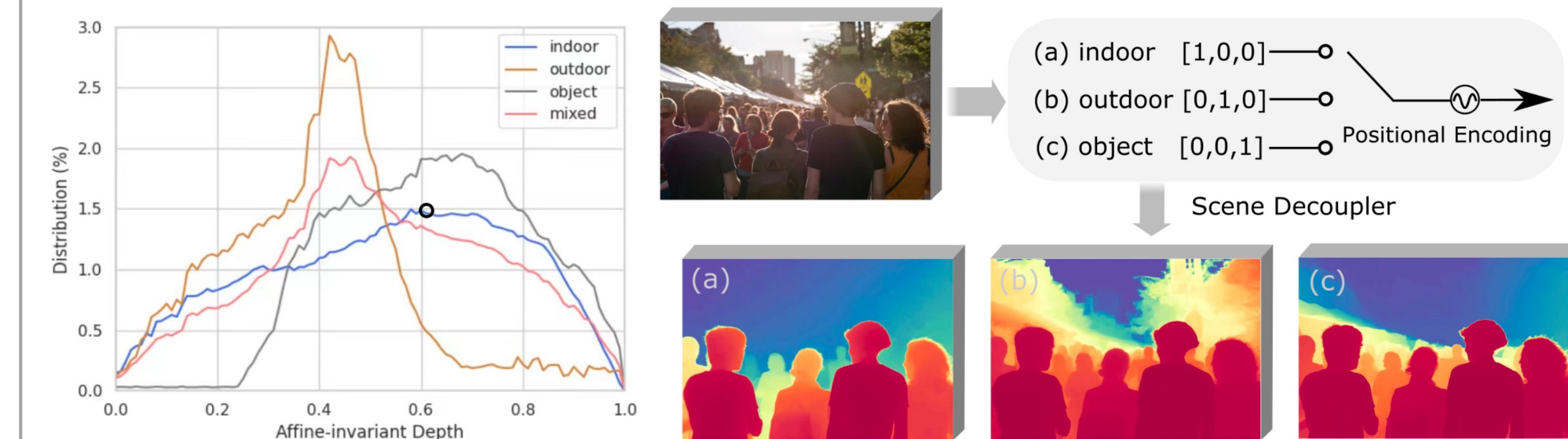


Contribution

(1) *Geometric Domain Switcher*: Ensure mutual guidance and geometric consistency

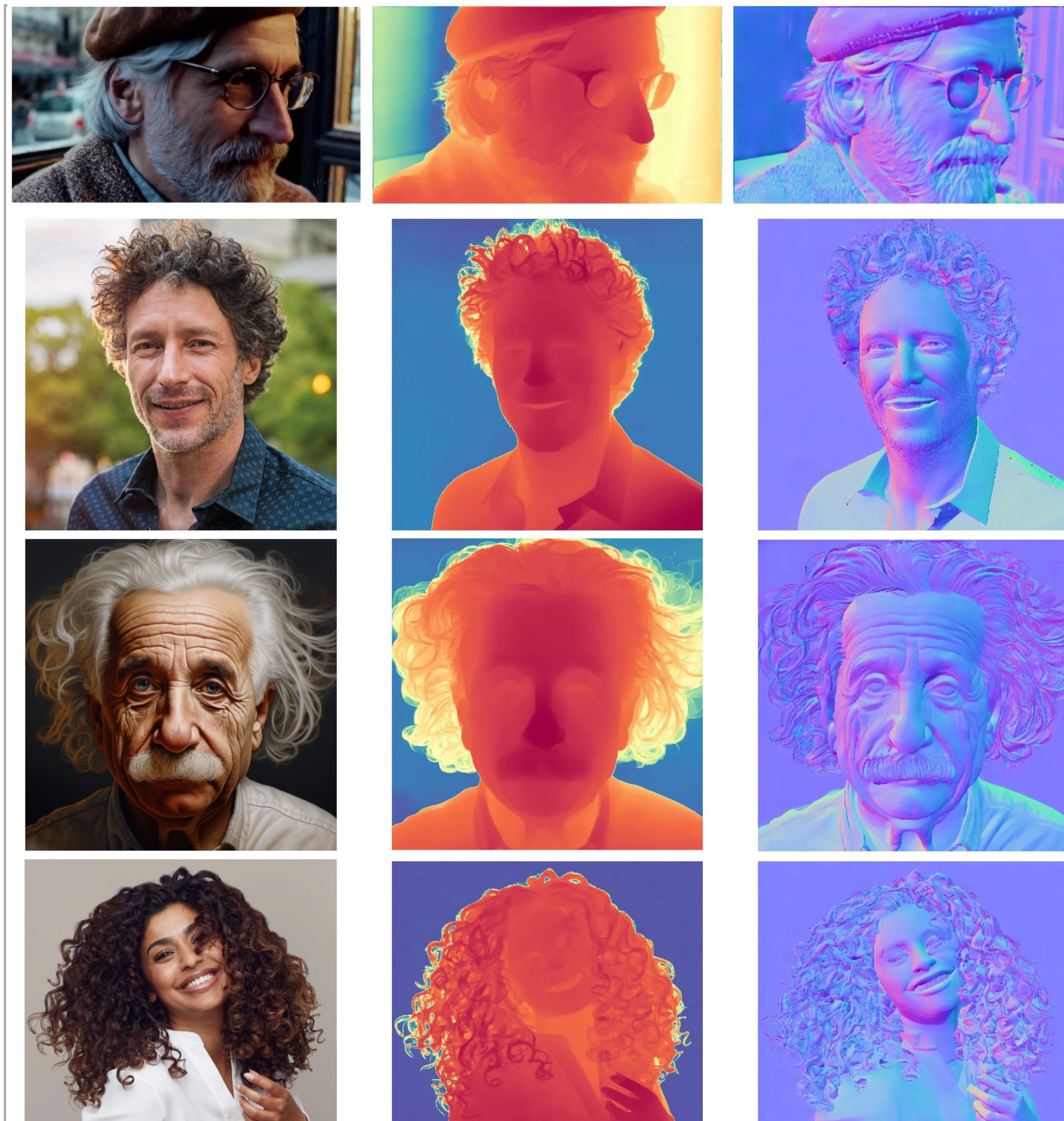


(2) *Scene Distribution Decoupler*: Recognize spatial layout (indoor/outdoor/object)

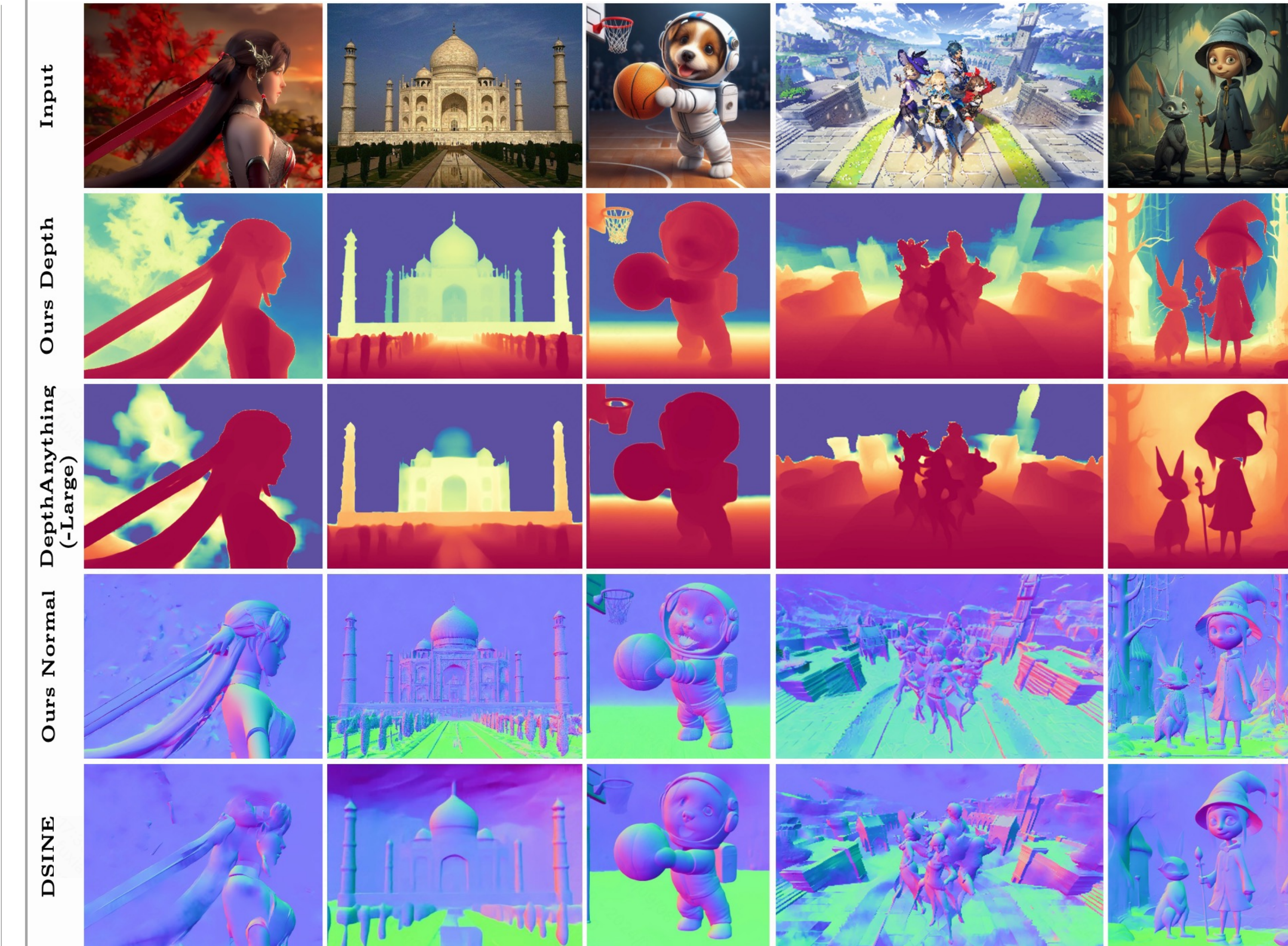


Method	Indoor			Outdoor			Object			Overall		
	AbsRel ↓	Mean ↓	GC ↓	AbsRel ↓	Mean ↓	GC ↓	AbsRel ↓	Mean ↓	GC ↓	AbsRel ↓	Mean ↓	GC ↓
Separate models	7.4	15.1	18.2	12.5	26.2	27.9	5.2	18.2	20.1	8.5	16.9	19.1
w/o Geometry Switcher	5.7	13.1	17.3	9.8	22.3	27.1	3.3	15.8	18.5	6.9	15.0	18.1
w/o Scene Decoupler	5.8	13.8	15.4	10.5	24.7	24.5	3.7	15.5	17.9	7.5	16.1	16.5
Full Model	5.5	12.6	14.7	9.6	22.1	23.5	3.5	15.4	17.6	6.7	14.8	16.2

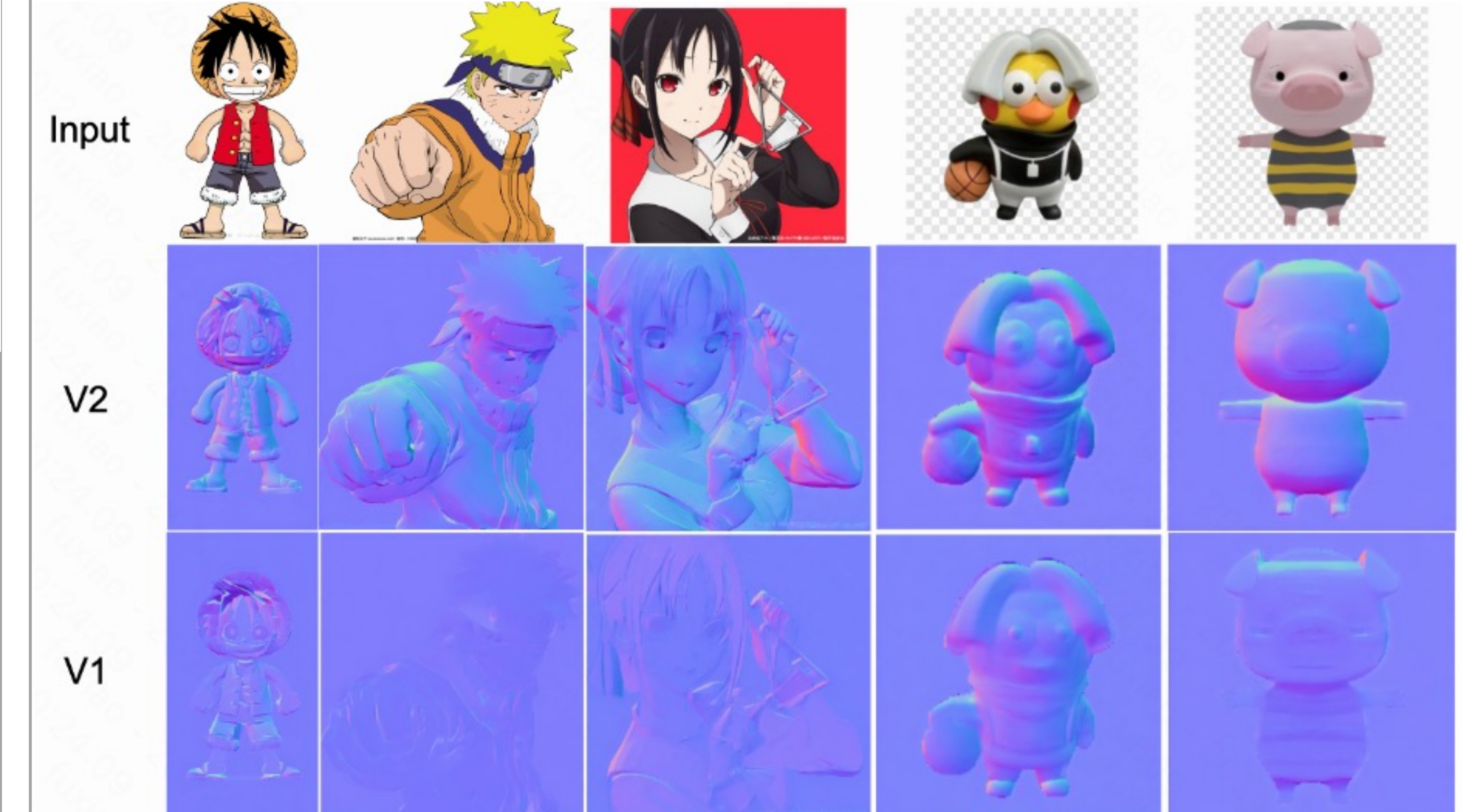
Hair-Level Details



Generalization Comparison



GeoWizard V2



Project Page



HF Demo



Code